Question 1 [6 marks]

Among BST and Heap, which one inserts slower? (a)Heap (b)BST (c) Neither (d)Both are the same.	[1]
What is the space complexity of out-place heapsort()? (a)O(1) (b)O(n) (c) O(n^2) (d)Will depend on the size.	[1]
Insert the following values into a BST and draw it: 45, 49, 53, 60, 10, 32, 27, 29	
The tree you drew for (3) is not balanced. a) True b)False?	
Delete the node 45 from the drawn BST and draw it again after deletion.	

Question 2 [6 marks]

Write a method named **kThMinimumPairSum** which takes **two integer arrays A and B**, and **an integer k** as parameters. The method should return the sum obtained by pairing elements from both arrays such that the smallest available element from **A** is paired with the smallest available element from **B**.

ASSUME: The **MinHeap** class is already implemented with **insert**() and **extract**() methods along with the necessary helper methods. You do not need to write the **MinHeap** class. Simply create an object of the class and use it accordingly.

NOTE: You must utilize a Min Heap to solve the problem. You are not allowed to generate all possible pair sums explicitly, Your solution time complexity should be less than O(N^2). You are not allowed to use any additional data structures except the heap.

Sample Input	Sample Output	Explanation
A = [4, 7, 2, 6] B = [9, 1, 3, 8] k = 3	14	1st pair → 2 + 1 = 3, 2nd pair → 4 + 3 = 7, 3rd pair → 6 + 8 = 14, 4th pair → 7 + 9 = 16, Here the 3rd smallest pair is 14 which is returned.
A = [3, -2, 7, 1, -5, 4] B = [6, -1, 2, -3, 8, 0] k = 4	5	1st pair → -5 + (-3) = -8, 2nd pair → -2 + (-1) = -3, 3rd pair → 1 + 0 = 1, 4th pair → 3 + 2 = 5 5th pair → 4 + 6 = 10, 6th pair → 7 + 8 = 15 Here the 4th smallest pair is 5 which is returned.

Python Notation	JAVA Notation

def kThMinPairSum(a, b, k):
    #to do
	public static int kThMinPairSum(int[] a, int[] b, int k){
	    //to do